Adapting and Benchmarking TerraMind Against Task-Specific Baselines for Multi-Modal Road Extraction across Urban, Peri-Urban and Rural Regions

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Abstract

Automating road extraction from open, 10 m-resolution remote sensing imagery is drastically more difficult than using high-resolution datasets ($< 1m$), which often focuses on urban areas, because of the thin typology of roads that can reach sub-pixel width. This leaves rural and peri-urban areas underserved, yet these areas remain a vital part of tasks such as disaster management and environmental monitoring. The development Geospatial Foundation Models (GFMs) offers the opportunity to better leverage multi-modal imagery from the open-access Sentinel 1 and 2 missions with a high revisit frequency of 5-days. As a result, a dataset using Sentinel-1/2 imagery and spanning urban, peri-urban and rural areas was curated to investigate using the multi-modal GFM TerraMind for road extraction under five decoder architectures, LoRA configurations across rank and target model, as well as benchmarking against the task-specific baselines, U-Net and D-LinkNet. The Fully convolutional Network (FCN) decoder under full fine-tuning was the strongest TerraMind adaptation (IoU=0.415). The vanilla LoRA setup struggled to capture the high "intrinsic rank" of road extraction and could not match full fine-tuning. Finally, the fully fine-tuned U-Net outperforms every TerraMind configuration, with only a quarter of the parameters in comparison (IoU=0.486). This gap is attributed to U-Net's multi-scale feature fusion that allows resolution of up to 20m to be preserved and leveraged in the decoder in contrast to the 160 m x 160 m resolution tokens inputted into the TerraMind encoder. The findings suggest that the adaptation approaches lack the resolution and representational capacity to match task-specific models on road extraction.

Keywords

Road Extraction, Semantic Segmentation, Remote Sensing

1 Introduction

Road extraction remains a complex, ongoing field of research relevant for routing during disaster relief, urban planning, and environmental monitoring, where a more specific example includes studying the relationship between economic development and road construction to inform environmental policy formations. The complexity of the task lies in the innately imbalanced class labels of road versus background, the need for topological connectivity, occlusion in the imagery. Currently, the automation of road extraction is an active field of research that commonly draws on the advancement in earth observation technology and deep learning methods.

Significant advancements have been achieved with high resolution ($5m$), RGB satellite imagery on benchmark datasets, such as Massachusetts [41] and DeepGlobe [6], which primarily focus on urban settings. However, the practical implementation of these methods are presently constrained by the cost of acquiring such imagery, associated restrictive licensing, and the poor generalisation to rural and peri-urban settings. The alternative approaches to address these constraints often leverage the global-coverage open data provided by the European Space Agency's (ESA) Copernicus Program [11]. Specifically, the Sentinel 1 and 2 missions of the program is considered state-of-the-art amongst publicly available, multispectral satellite imagery as it provides imagery with up to 10m in resolution, and 5-day revisiting rate. These satellite images combined with labels obtained from open-access platforms, such as OpenStreetMap, offer a practical research direction for addressing the constraints of the mainstream road extraction research.

Furthermore, the abundance of open-access satellite data enabled the development of Geospatial Foundation Models (GFM) - the subset of large vision models pre-trained on massive amounts of unlabelled satellite data. GFMs aim to acquire remote sensing imagery context during pre-training to be utilised for various earth observation tasks within zero-shot or few-shot adaptation [23]. TerraMind [22] is a multimodal GFM that has demonstrated state-of-the-art fine-tuning performance [22, 40] on the PANGAEA benchmark [39], which includes 11 Earth observation tasks. However, full fine-tuning can potentially lead to the erasure of pertinent pre-trained features, and, for increasingly large models, the method is limited in scalability due to increasing cost and computational resources required. Hence, Parameter-Efficient Fine-tuning techniques (PEFT) focus on updating a subset of the model parameters during fine-tuning, which significantly reduces resource demands [16]. Amongst this class of techniques, Low-Rank Adaptation (LoRA) has shown potential for mitigating catastrophic forgetting [9].

2 Research aim and objectives

This project aims to investigate whether adapting a GFM, TerraMind, in terms of decoder selection and the LoRA fine-tuning approach, can match the task-specific, medium-sized U-Net and D-LinkNet models for road extraction through segmentation in the urban, peri-urban, and rural South African context with open-access medium-resolution, multispectral satellite data. Specifically, the aim is broken down into the following research questions: (1) Which of the the five decoders investigated maximises TerraMind's road segmentation performance? (2) By varying LoRA's target module and rank parameters, to what extent can LoRA recover the performance of full fine-tuning? and (3) How does TerraMind's

best-performing variant - obtained from the above experiments - differ in performance relative to the medium-sized models (U-Net and D-LinkNet) across the performance metrics?

3 Background and Related Works

Research on road extraction is most active with high resolution, RGB, urban satellite data. Benchmark datasets, such as Massachusetts [41] and SpaceNet [10], that proliferated the research on deep learning for earth observation centre on urban and peri-urban environments. However, the construction of new roads often occur in more rural areas. Therefore, the need to accurately extract roads in urban, peri-urban, and rural settings becomes vital in obtaining sound information on road distribution, especially for government decision making[24]. Additionally, the topology of roads are defined by their narrow, continuous tubular structure, and their width is often less than 10m, especially with rural [42] or residential roads. As a result, road extraction research mainly used high resolution (1m-5m) or very-high resolution (VHR) (<1m) remote sensing images, often limited to RGB bands, to extract road networks [24]. However, practical applications of these methods are prohibited by their expensive cost, low revisit period, and small coverage [24].

Hence, the following subsections describe the background and related works in terms of opportunities with open-access satellite data, deep learning methods and in particular with larger models as well as the relevant adaptations associated with them to make automatic road extraction for up-to-date road maps more feasible.

3.1 Satellite Data

The Sentinel 1 and 2 missions of the Copernicus Program [11] provides publicly-accessible, global-coverage, multispectral satellite data captured at a 5-day temporal frequency presents the opportunity to automate road extraction at scale. This is particularly relevant for use cases that demand timely updates, such as monitoring road development to study the associated environmental impacts[49].

Sentinel-2 captures multispectral imagery with 10 m - 60 m resolution bands - including RGB, red Edge, Near Infrared (NIR), short-wave Infrared (SWIR) [25]. Bands beyond the RGB visible spectrum are useful in distinguishing surfaces such as vegetation, and water[51]. Sentinel-1 captures imagery with Synthetic Aperture Radar sensors, which can penetrate canopy and cloud cover that commonly blocks Sentinel-2 sensors [10, 43]. Although road extraction research predominantly based on RGB channels [42], works of [2, 34] demonstrated that fusing SAR and multispectral imagery improves in road segmentation results.

3.2 Deep Learning Methods

Amongst the various deep learning methodologies applied to road extraction, U-Net [47] remains state-of-the-art and a common baseline. The architecture is characterised by symmetric encoder-decoder structure with skip connections. The encoder extracts features, the decoder produces task specific predictions, while the skip connection achieves multi-scale feature fusion that preserves feature details. Another common baseline is D-LinkNet [60], winner of the CVPR DeepGlobe 2018 Road Extraction Challenge. Unlike the U-Net, this model is specifically designed for road extraction, and utilises dilated convolutions in the central block to expand the receptive field.

Often, the ResNet [18] model, a convolutional neural network (CNN) encoder pre-trained on ImageNet [7], is used as a backbone in medium-sized models for transfer learning, faster convergence, and the residual connections that combat the vanishing gradient problem [51]. For road extraction, ResNet34 has achieved notable results as a backbone in the U-Net [4] and D-LinkNet [60] architecture.

The success of large models, known as foundation models, in natural language processing has encouraged scaling model training in terms of parameter count, dataset size, and compute resources [27, 40] across various deep learning application. This includes road extraction, where the the Segment Anything model (SAM) [29] and SAM 2 [46] are most frequently subject of adaptation [13, 14, 28, 32, 37, 56]. These models are vision transformer models developed by Meta AI, and pre-trained on natural RGB images. The accessibility of natural-image-based deep-learning segmentation methods, such as the SAM models [29, 46] and ResNet [18], as well as RGB benchmark datasets [6, 41] likely contributed to the prevalence of RGB-focused methodologies, despite the potential of incorporating multispectral and multimodal data [2, 34]. When channels beyond RGB are used, common approaches include increasing the number of channels in the network input layer, or to use image fusion techniques [51]. Although input channel expansion is effective, the additional channels do not significantly improve upon RGB training results because modification of the pre-trained model structure can cause the original weight values to fail [51]. For foundation models such as SAM, the domain gap between natural and satellite data needs to be addressed before applications to Earth observation [52].

In this regard, the recent emergence of Geospatial Foundation models (GFMs), pre-trained on volumes of satellite data, offer the opportunity to better leverage open data satellite imagery and multi-modal information. TerraMind [22], displayed in Figure 2, is a multimodal GFM pre-trained on Sentinel satellite imagery, and uniquely employs dual scale feature representation at the token and pixel level. The token level captures high-level cross-modal correlation, while the pixel level preserves spatial nuances critical for segmenting thin structures such as roads. Furthermore, TerraMind achieved state-of-the-art fine-tuning performance on PANGAEA [39], a global-coverage earth observation benchmark for GFMs. Importantly, the researchers noted that it was the only GFM approach to outperform task-specific U-Net [22].

3.3 Adaptation

Although TerraMind exhibits promising zero-shot performance on water body mapping [22], adaptation remains essential to account for the unique narrow tubular topology of roads.

As models scale, the computational resource and cost associated with retraining all model parameters, also known as full fine-tuning, becomes less feasible. As a solution, parameter-efficient fine-tuning (PEFT) alleviates compute and memory demands by only updating a subset of the model's parameters [9]. LoRA [19] is a common PEFT approach, visualised in Figure 4 that has been shown to match or outperform full fine-tuning results [9, 55]. The approach introduces

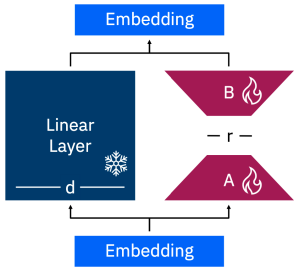


Figure 1: LoRA reparameterisation of the frozen weights $W \in \mathbb{R}^{d\times d}$ (dark blue) with trainable low rank matrices $A \in \mathbb{R}^{r\times d}$ and $B \in \mathbb{R}^{d\times r}$ (red) using r to denote the bottleneck dimension and embeddings (lighter blue). Sourced from [9]

a pair of trainable low-rank matrices, A and B, scaled by a factor, γ, into the target module of each transformer block in the model. The product of these two matrices after fine-tuning forms the update matrix used to reparameterise the original Transformer model weight matrix. Thus, LoRA reduces the trainable parameters of a pre-trained weight matrix, $W \in \mathbb{R}^{d\times d}$, from d^2 to $2 \cdot d \cdot r$, where r is the bottleneck dimension. Performance benefits can include mitigation of overfitting [32], and catastrophic forgetting [9]. Moreover, only the low-rank matrices need to be stored per task to reparameterise the frozen backbone, making it more space efficient when fine-tuning the same model for multiple downstream tasks. In road extraction, LoRA has predominantly used to adapt SAM [32, 37]. [35]

Apart from fine-tuning, adaptation can also include modifying architectural choices[9]. TerraMind's pre-trained transformer decoder was designed for generation tasks rather than segmentation tasks. Hence, modifying the decoder architecture is necessary to adapt TerraMind to the dense prediction task required for road segmentation. Additionally, model performance depends not only on the extracted encoder features, but also the decoder's ability to leverage them.

4 Methodology

This section describes the design and setup of the experiments. Specifically, the following subsections explain and motivate the TerraMind adaptation framework, baseline architectures, evaluation metrics, dataset curation, and the training protocol across the experiments.

4.1 Framework Design

The experiments used the TerraTorch [15] fine-tuning framework and is divided into three parts: decoder selection, LoRA configuration analysis, and baseline benchmarking. The final part evaluated the baseline models against the best-performing TerraMind adaptation.

4.1.1 Decoders. Using full fine-tuning, five decoder architectures were evaluated for road extraction: a Linear decoder decoder, Fully Convolutional Network (FCN) [36], UPerNet [53], U-Net [47], and DSCNet [44].

The Linear decoder decoder is a transposed convolution without nonlinearity. Its simplicity is well-suited for evaluating the quality of the encoder features. The FCN decoder [36] reconstructs spatial resolution through convolution blocks, each sequentially applying a transposed convolution, a 3 x 3 convolution, Layer Normalization, and GeLu activation. Building upon this, the UPerNet decoder [53] uses a Feature Pyramid Network (FPN) [33] for multi-scale features and a Pyramid Pooling Module (PPM) [58] to capture global context. Instead of using a FPN, the U-Net incorporates multi-scale features through skip connections, and uses decoder blocks containing interpolation, concatenation, as well as convolution layers followed by batch normalisation, and ReLu activation. Finally, the DSCNet [44] decoder utilises a sequence of blocks containing upsampling, followed by a multi-view, multi-scale feature fusion to reconstruct spatial resolution with geometric inductive bias specialised for the preservation of the continuity in curves of elongated structures. The fusion is achieved through (1) placing each multi-scale feature through three convolutions - a standard convolution and two deformable snake convolutions, deforming along the x-axis and y-axis, respectively - to obtain the views; and (2) placing the concatenated views through another standard convolution.

The DSCNet decoder was not available in TerraTorch like the other decoders. Instead, the model code was extracted from [8] and adapted for registration with TerraTorch. This included decoupling the decoder from the original model, distributing the interpolated encoder features to their respective blocks, and adding an argument to take in `out_channel` specifications like the other multi-scale decoders.

4.1.2 LoRA. The effect of target module and rank, r, were investigated using TerraMind fitted with the Linear decoder before reporting the best configuration that balances performance with parameter efficiency. The target module setups include attention only, multi-layer perceptron (MLP) only, and all transformer modules. Thereafter, TerraMind LoRA at r = {4, 16, 64 } were evaluated. The log scaling was chosen as LoRA's low "intrinsic rank" means increasing rank arbitrarily does not cover a more meaningful subspace and yields diminishing returns [19].

4.1.3 Architecture. Figure 2 displays the simplified visualisation of the TerraMind architecture used in this project. The input bands include the VV and VH bands from the Sentinel-1 GRD data product as well as the RGB, Red Edge, NIR and SWIR bands from the Sentinel-2 L2A data product. In the TerraMind model, each modality is separately split into into 16 x1 6 patches, giving 256 tokens per modality. Each patch is linearly projected into 768 dimension vectors as tokens and concatenated together for the backbone. The TerraMind ViT backbone used consists of 12 transformer layers and each layer outputs tokens. The tokens from the selected layers are reshaped into images. The FCN and Linear Decoder only need the tokens from the last transformer layer. In contrast, the three multi-scale decoders take tokens from four transformer layer, per [21], to be reshaped and then resized with the *LearnedInterpolateToPyramidal* neck to build the hierarchical features. Finally, the decoder outputs feature maps, which the decoder head uses to produce the class logits and predicted road mask.

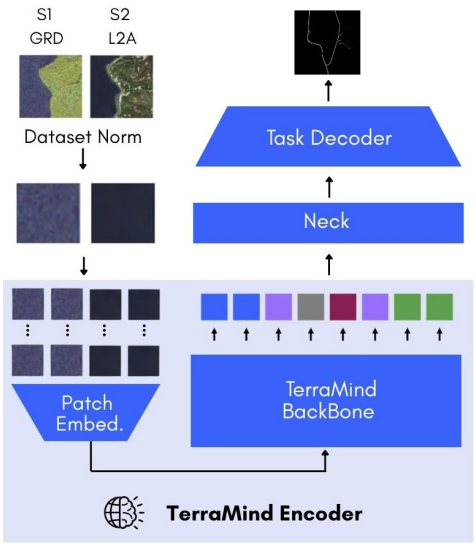


Figure 2: TerraMind architecture components in adaptation pipeline: 256×256 normalised input, per-modality 16 × 16 patch projected to 768-d vector embedding and covers 160 m x 160 m area, 12-layer ViT, neck (SelectIndices, ReshapeTokensToImage to 768 × 16 × 16, and, for multi-scale decoders, LearnedInterpolateToPyramidal), and task decoder

4.1.4 Baselines. The adaptations of TerraMind were evaluated against the medium-sized models U-Net [47] and D-LinkNet [60], with ResNet34[18] as their backbone. The U-Net was natively supported by the TerraTorch framework, which used the implementation from the Segmentation Models.PyTorch (SMP) library.

In contrast, D-LinkNet required integration. ResNet34 [18] and the original DeepGlobe road extraction challenge [6] is based on RGB imagery. Hence, the first convolution of D-LinkNet was patched using the SMP library, the same way as U-Net, to adapt the D-LinkNet model extracted from GitHub [59] for multi-band input. D-LinkNet's direct probability output was also removed to output raw logits. Finally, a wrapper class was created to register D-LinkNet with TerraTorch.

4.2 Dataset

To the best of our knowledge, there is no road extraction dataset that (1) uses open-access satellite imagery, (2) incorporates both multispectral and SAR channels, as well as (3) contains diverse levels of urbanisation. Hence, we curated a dataset, named Roads of South Africa (ROSA), that covers urban, peri-urban, and rural roads across the 9 biomes of South Africa, for a diverse representation of the region.

We utilised Google Earth Engine to obtain and preprocess satellite imagery. The specific bands obtained include the RGB, red Edge, Near Infrared (NIR), short-wave Infrared (SWIR) multispectral bands from Sentinel-2; and the radar bands VV and VH from Sentinel-1. Preprocessing was achieved with [61]'s open-source data generation JavaScript. which for the multispectral bands include cloud and shadow filtering; and across all bands include noise reduction through temporal aggregation, co-alignment at 10m resolution, and concatenation into a single multi-band image stack. The vector road labels were obtained from the South African government's Chief Directorate National Geo-Spatial Information (CDNGI) portal [31], with footpath and street excluded. We also manually inspected the test split labels for any outdated or incomplete labels.

Dataset stratification across urbanisation and biomes was achieved through a semi-automated process, with information obtained from the ESA's World Settlement Footprint [12] and South Africa's 2025 National Vegetation Map [50] respectively. Across the biomes, we manually identified urban regions, while rural and peri-urban regions were randomly sampled by a fellow team member. The same team member also rasterised the road labels, computed the normalisation statistic of the training split, derived other relevant metadata, and added enhanced RGB imagery as three additional channels.

4.3 Evaluation

Model performance was measured through pixel-based metrics and a graph-based metric. The evaluation was conducted on the entire test split of the ROSA Dataset as well as separately across the three levels of urbanisation.

The standard pixel-based metrics used in road extraction include Precision, Recall, F1-score, and Intersection over Union (IoU) [38]. In particular, F1-score and IoU account for the class imbalance between road and background pixels to provide a more representative evaluation. For Equations 1 to 4, true positives (TP) are the number of correctly classified road pixels, false positives (FP) are the background pixels misclassified as roads, and false negatives (FN) are the road pixels misclassified as background.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{1}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{4}$$

Additionally, the pixel-based metric centerlineDice (clDice) [48] was used to assess connectivity in segmentation prediction. By measuring the intersection of road centre line and the full segmentation masks, clDice penalises structural breakages more heavily than standard pixel metrics.

The topological correctness of predicted graph networks is also measured with the graph-based metric [38], Average Path Length Similarity (APLS) [10], as standard pixel-based metrics fail to assess road continuity [10]. However, converting raster mask predictions to graph representations can introduce noise that systematically inflate or deflate the metric. This systematic bias is accepted as a trade-off for the insight provided by the APLS.

For all the above metrics, higher scores indicate better performance.

Besides road extraction performance, overfitting is measured with relative generalisation percentage defined in Equation 5, with higher values represent worse overfitting and $IoU_{evaluation}$ referring to either IoU derived on the validation or test split.

$$\text{RG } \% = \frac{IoU_{train} - IoU_{evaluation}}{IoU_{train}} \quad (5)$$

4.4 Experimental Setup

Across the experiments, the baseline model's ResNet34 backbone contains approximately 21M parameters, while the TerraMind adaptations used the TerraMind 1.0 Base backbone that contains 12 transformer blocks and has a total of 80M parameters . Except for the TerraMind Linear decoder configurations, all other decoder's were configured such that the total parameters excluding the backbone amounts roughly to 20M, in reference to [9].

The geometric transformations - rotation, reflection, and flipping - were applied with the D4 method from the Albumentations library. Additionally, another teammate's implementation of random cropping in the training set, and sliding window for the test and validation set was used. The tiles were cropped at the fixed size of 256 x 256 from the original 512 x 512 image to not reduce spatial resolution, which is crucial for performance [30]. Colour augmentations were avoided to prevent loss of the spectral information that is important for remote sensing image interpretation [1, 17].

Preliminary experiments, Appendix A, guided the selection of normalization, RGB channel enhancements, and precision settings. Consequently, dataset-specific normalisation statistics were adopted over TerraMind's pre-training defaults across all twelve original bands. Additionally, bf16-mixed precision was used to minimize training overhead.

Dice coefficient loss, a region-based loss function, was used in all the experiments because [54] found that region-based loss functions consistently outperform distribution-based alternatives, including Binary Cross-Entropy (BCE), in F1 score and IoU. Researchers suggest that this advantage stems from the loss function's global evaluation approach, in contrast to the pixel-wise evaluation of distribution-based losses, and ability to mitigate the severe class imbalance inherent to road extraction [42].

The HPO of learning rate follows the Geo-Bench fine-tuning procedure [30]. Specifically, all models trained in this project was selected via Bayesian optimisation using TerraTorch Iterate's [20] Optuna study integration. The study evaluated 16 trials of up to 20 epochs each, with Hyperband early pruning enabled. For the LoRA experiments, learning rate HPO was conducted with rank fixed at 16 for each of the three target module setups. To avoid the need of re-optimising learning rate across LoRA rank experiments, the scaling factor, γ, was fixed at $1/r$, and the low-rank matrices A and B were initialised randomly and to zero, respectively [5]. Additionally, only one global learning rate was tuned for each model due to resource constraints.

The setup choices made in reference to [9] are as follows. The batch size was set to 8 for a faster convergence on our relatively small dataset [9]. Each model trained for up to 100 epochs with early stopping after 20 epochs without improvement. The AdamW optimizer was used with default pytorch.optim parameters and ReduceLROnPlateau with $patience = 4 epochs$ and $factor = 0.5$.

Additionally, all experiments and HPO were conducted on a single NVIDIA L40s GPU. To estimate training variance of the models, each reported configuration is trained with seeds 0, 1, and 2, with results reported at mean ± std in Section 5.

5 Experiments and Results

In this section, the results from the decoder and LoRA experiments are first discussed individually. Thereafter, the best TerraMind configuration across those experiments is benchmarked against the baselines and discussed.

5.1 Decoder

Table 1 and Figure 3 displays the empirical performance and sample prediction of TerraMind with different decoders, respectively. Notably, the FCN decoder exhibited the strongest performance across most metrics.

The decoders with multi-scale feature fusion - the U-Net, UPerNet, and DSCNet decoders - exhibited weaker performance across most pixel-based metrics compared to the single-resolution decoders - the Linear decoder and FCN decoders. In particular, the DSCNet decoder that fused features from multiple scales and views performed the worst across the pixel-based metrics. When inspecting the outputted prediction masks, the road edges of the single-resolution decoders appear more uniform in width and tubular compared to the other decoders, with a lumpier road edge and thicker roads that blend together. The observation is particularly noticeable in denser road networks, as shown in Figure 3. Although multi-scale feature decoders were designed to preserve spatial resolution, the finest resolution accessible in the Transformer encoder layers remain at 160 m x 160 m, as demonstrated in Figure 2. Additionally, ViT backbones do not provide varying level of spatial abstraction that the multi-scale decoders and maintain more uniform representation of features[45]. When scaling the features to the expected sizes of the multi-scale decoders, the interpolated representation may have also increased the difficulty in discerning the true road width from the features.

Notably, compared to other decoders, the Linear decoder, with only a single transposed convolution for upsampling, exhibits a relatively strong amongst the pixel-based metrics, specifically superior in clDice, while exhibits the worst performance on the graph-based metric APLS. This is reflected visually in Figure 3 with more speckled and disjoined segmentations. The Linear decoder decoder also exhibits the strongest presence of the grid lines as an artefact of the patched input, which are non-overlapping in nature, ingested by the transformer. The dichotomous performance on the clDice versus the APLS, respectively, is indicative of the TerraMind encoder's inherent ability to detect linear structures like roads, and the need of specialised mechanisms to enforce important road-specific characteristics such as continuity. This divergent performance within the Linear decoder metrics also suggests that while deeper decoder levels removed the speckled noise from the Linear decoder predictions and improved the continuity of roads through gradual upsampling, the same process also erased the roads predicted as wispy segments.

The FCN decoder partially eliminates speckled and disjointed artifacts by using convolutional layers to enforce spatial continuity and enhance connectivity, avoiding the multi-scale feature aggregation that obscures road width.

Table 1: Comparative full fine-tuning performance of decoder architecture choice with TerraMind V1 Base backbone on the ROSA dataset test split

Decoder	Precision	Recall	F1	IoU	clDice	APLS
Linear	0.5684 ± 0.0046	0.5811 ± 0.0056	0.5747 ± 0.0017	0.4032 ± 0.0016	**0.4898 ± 0.0031**	0.2909 ± 0.0012
FCN	**0.5868 ± 0.0073**	0.5860 ± 0.0032	**0.5864 ± 0.0052**	**0.4149 ± 0.0052**	0.4784 ± 0.0015	**0.3920 ± 0.0052**
U-Net	0.5272 ± 0.0029	**0.5882 ± 0.0028**	0.5560 ± 0.0012	0.3851 ± 0.0011	0.4554 ± 0.0015	0.3537 ± 0.0062
UPerNet	0.5228 ± 0.0042	0.5838 ± 0.0125	0.5516 ± 0.0040	0.3808 ± 0.0038	0.4630 ± 0.0027	0.3313 ± 0.0030
DSCNet	0.4907 ± 0.0052	0.5739 ± 0.0028	0.5291 ± 0.0029	0.3597 ± 0.0027	0.4461 ± 0.0023	0.3239 ± 0.0060

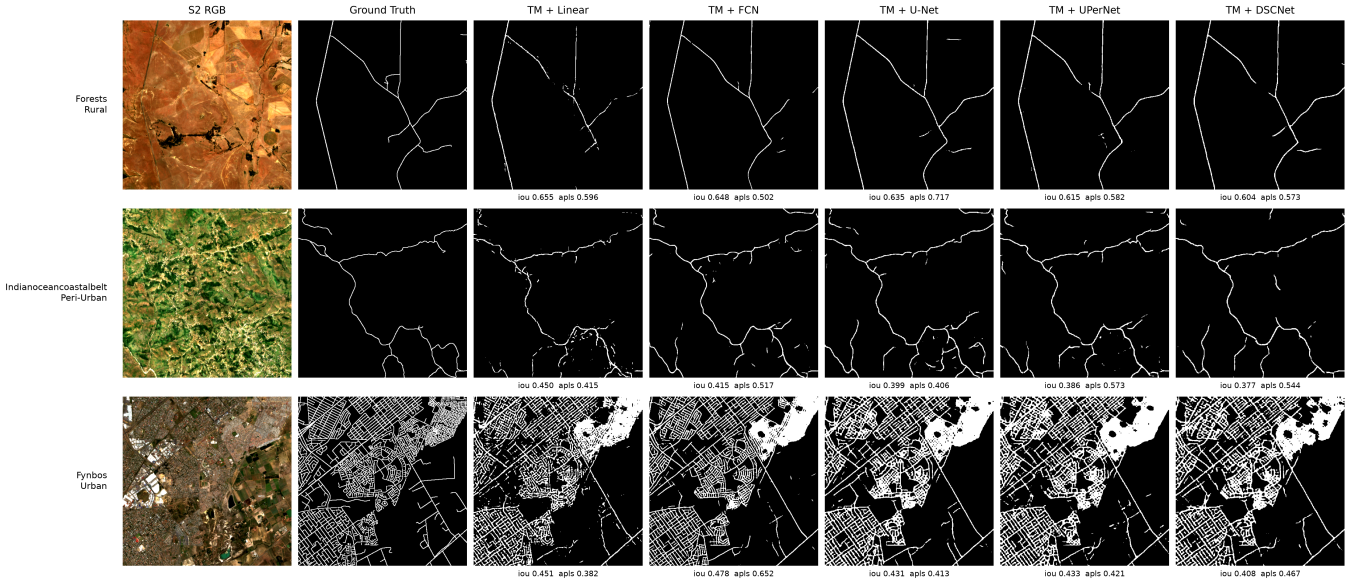


Figure 3: Per tile mean empirical performance across the three seeds and predictions masks from ROSA dataset test split of TerraMind with different decoders.

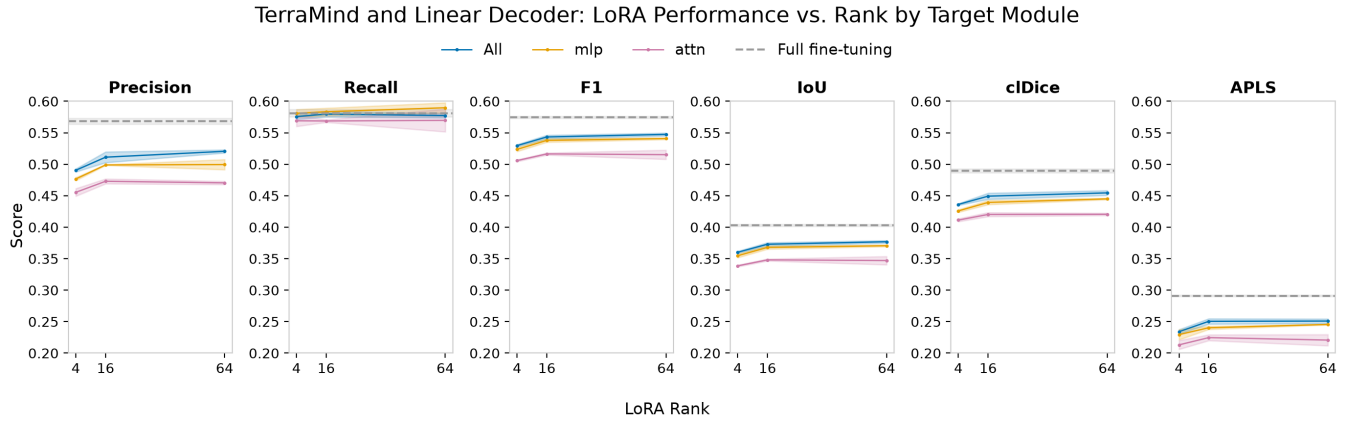


Figure 4: Comparative performance the LoRA configurations against full fine-tuning with TerraMind V1 Base Backbone fitted with the Linear decoder on the ROSA dataset test split

Table 2: Comparative performance of LoRA across target modules, all , MLP and attention (attn) with TerraMind base-sized backbone on the ROSA dataset test split with $\gamma = 1/r$

Target	Rank	Precision	Recall	F1	IoU	clDice	APLS
All	4	0.4904 ± 0.0024	0.5757 ± 0.0044	0.5296 ± 0.0018	0.3602 ± 0.0017	0.4362 ± 0.0013	0.2340 ± 0.0034
All	16	0.5112 ± 0.0085	0.5797 ± 0.0044	0.5433 ± 0.0029	0.3729 ± 0.0027	0.4493 ± 0.0049	0.2505 ± 0.0043
All	64	**0.5205 ± 0.0025**	0.5773 ± 0.0036	**0.5474 ± 0.0023**	**0.3769 ± 0.0021**	**0.4545 ± 0.0038**	**0.2509 ± 0.0036**
MLP	4	0.4766 ± 0.0025	0.5810 ± 0.0061	0.5236 ± 0.0032	0.3547 ± 0.0030	0.4259 ± 0.0021	0.2294 ± 0.0081
MLP	16	0.4989 ± 0.0006	0.5834 ± 0.0058	0.5379 ± 0.0028	0.3679 ± 0.0026	0.4393 ± 0.0033	0.2403 ± 0.0025
MLP	64	0.4995 ± 0.0081	**0.5897 ± 0.0080**	0.5408 ± 0.0015	0.3706 ± 0.0014	0.4450 ± 0.0012	0.2455 ± 0.0006
attn	4	0.4554 ± 0.0060	0.5690 ± 0.0089	0.5058 ± 0.0015	0.3386 ± 0.0014	0.4113 ± 0.0023	0.2131 ± 0.0070
attn	16	0.4729 ± 0.0039	0.5687 ± 0.0020	0.5164 ± 0.0017	0.3481 ± 0.0016	0.4202 ± 0.0032	0.2246 ± 0.0046
attn	64	0.4704 ± 0.0026	0.5697 ± 0.0181	0.5152 ± 0.0073	0.3470 ± 0.0066	0.4205 ± 0.0019	0.2207 ± 0.0089

5.2 LoRA

While Figure 4 and Table 2 demonstrates that LoRA matches full fine-tuning performance in recall, all tested LoRA configurations for the TerraMind encoder underperform across the other metrics. Although LoRA has shown to match or exceed full fine-tuning performance in some geospatial tasks [9, 55], these results suggest that the current LoRA configurations lack the representational capacity necessary to adapt TerraMind for road extraction. Table 3 shows that LoRA's IoU_{train} at 0.3392 is significantly lower than the full fine-tuning IoU_{train} at 0.4966, reflecting a failure to recover the representational capacity of full fine-tuning.

Additionally, recall of LoRA (target module = all and $\gamma = 1/64$) matches full fine-tuning while precision decreases by roughly nine percent. This indicates that the additional errors are false road predictions rather than missed detections of road. The performance decrease widens with the topological sensitivity of the metric F1, clDice, and APLS, respectively. Since the process of calculating clDice and APLS both involve skeletonizing the road masks, they are largely insensitive to over-prediction of width. Thicker road mask predictions alone would not produce this ordering. Therefore, the errors are more consistent with spurious road-like structures and with widened masks that merge adjacent features, both of which introduce skeleton branches outside the ground truth.

Part of the performance gaps, identified above, may be architectural. TerraMind's input passes through a patch-wise linear projection, which converts the pixel-level input patches to the patch embeddings in a lower-dimension latent space. These are frozen during LoRA, and only adapted during full fine-tuning. Although TerraMind's pretraining preserves fine spatial nuances, such as linear structures, in its pixel-level patch embeddings [22], it does not explicitly differentiate roads from other linear features, leading to the observed drop in precision.

Compared to full fine-tuning, standard LoRA exhibits constrained expressiveness due to key structural limitations. First, assigning a fixed rank across all Transformer layers [19] fails to account for varying layer-wise adaptation requirements [57]. Second, when weight updates are decomposed into direction and magnitude components, LoRA tends to couple magnitude and directional changes, which limiting its ability to make fine-grained parameter adjustments [35]. Full fine-tuning does not exhibit the limitation above. As a result, these limitations potentially bounded the achievable update of the LoRA approach.

In Figure 4 and Table 2, target module selection is shown to primarily drive performance across all tested LoRA configurations. LoRA applied to the MLP modules alone captures most of the achievable performance. On the other hand, adding attention modules alongside the MLP modules improves precision, but offers negligible overall gains across F1, IoU, clDice, and APLS. While this aligns with findings that MLP layers exhibit higher effective rank [3], the advantage is partly parameter-driven, as MLP adapters carry roughly 1.7× the parameters of attention adapters.

The diminishing returns beyond rank 16 are interpreted as a symptom of the LoRA scaling factor rather than as evidence of a low intrinsic rank for the task. The original lora paper [19] reported that low ranks of 4, 8 and 16 were sufficient as negligible gains were observed at rank 64. However, [26] demonstrated that the standard scaling factor $\gamma = \alpha/r$, with fixed α, causes gradient collapse where the learning rate is slowed until larger ranks perform no differently from small ones. They further suggest that [19]'s conclusion is an artefact of this scaling. The results shown in Figure 4 reflects the same pattern of collapse, where ranks beyond 16 at $\gamma = 1/r$ yields minimal differences across the six metrics. Although the scaling factor $\gamma = 1/r$ was chosen for its ability to decouple learning rate HPO and rank sweeps [5], it is suboptimal for road extraction as the results suggest that it cannot exploit higher rank updates.

5.2.1 Generalisation. Although LoRA underperforms full fine-tuning, LoRA's average IoU_{test} performance exceeds its average IoU_{train} performance, calculated from the final training epoch across three seeds, and exhibits a negative relative generalisation gap. Table 3 shows that full fine-tuning overfits the training, unlike the LoRA approach. Applying LoRA achieves a regularisation effect due to the limited trainable parameters, which help preserve pretrained features. However, the regularisation effect of LoRA does not compensate for the deficit observed in the performance. As a result of the overall underperformance of LoRA, the LoRA runs are excluded from the best configurations section.

5.3 Best configurations

In Table 4, the baselines, U-Net [47] and D-LinkNet, are compared against the best TerraMind configuration, which the fully fine-tuned TerraMind with a FCN decoder. Example predictions of these models are displayed in Figure 5 The U-Net consistently exhibits the strongest performance in the pixel-based metrics, and only falls

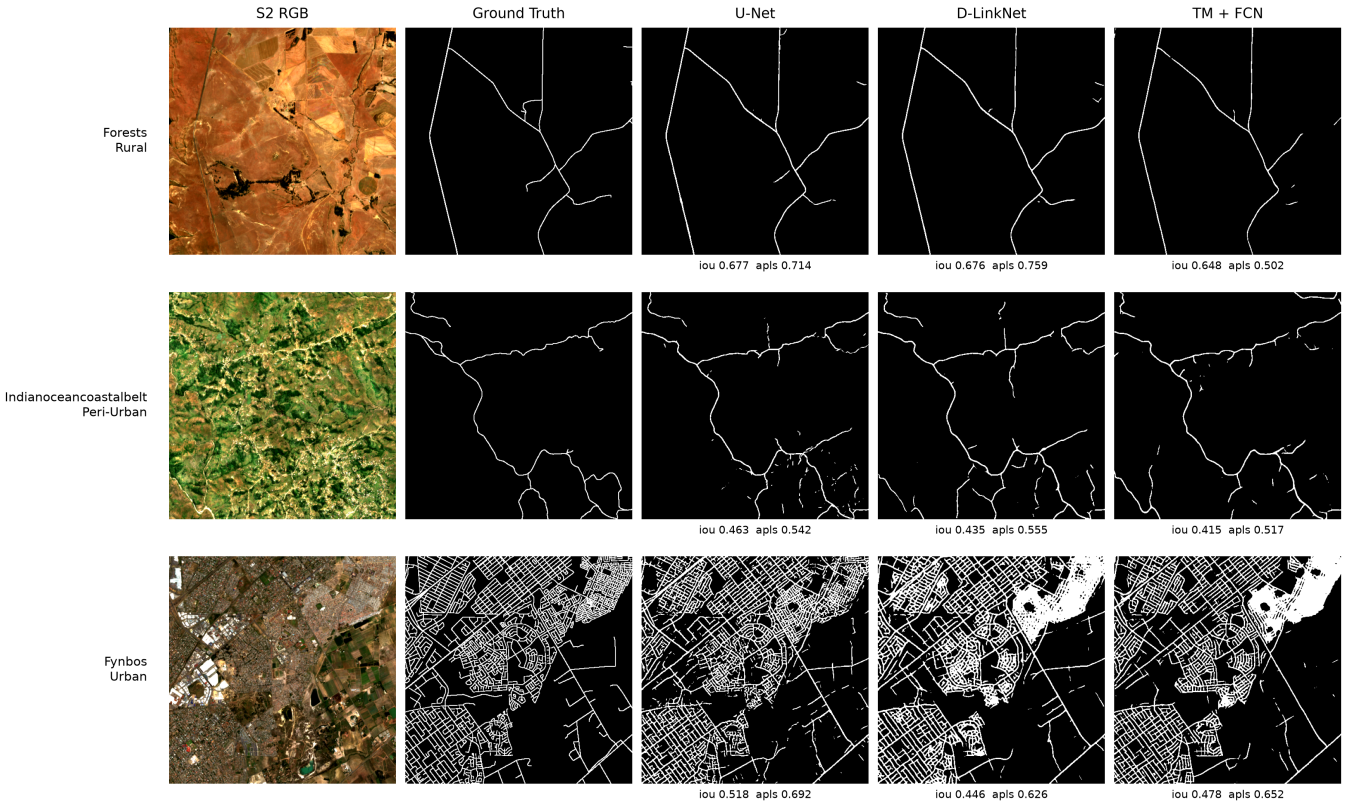


Figure 5: Sample predictions grid of baselines, U-Net and D-LinkNet, with the best TerraMind adaptation (FCN decoder using full fine-tuning)

Table 3: Relative generalisation percentage on test set of TerraMind + Linear decoder versus LoRA with target modules = all, $r = 16$, $\gamma = 1/r$

Fine-tuning	IoU_{test}	IoU_{train}	RG %
Full	0.4032 ± 0.0016	0.4966 ± 0.0095	18.808
LoRA	0.3729 ± 0.0027	0.3392 ± 0.0026	-9.037

slightly behind D-LinkNet's average APLS, though the standard deviations suggest that U-Net can outperform D-LinkNet marginally. U-Net is also the only model where dense urban road networks do not merge together to form blobs, as shown in Figure 5 This observation may be the result of D-LinkNet's coarser addition-based feature fusion method, and dilated convolution as a centre block, respectively. Specifically, the dilated convolution increases the effective receptive field to incorporate long-range context, which improves road connectivity. However, unlike U-Net's concatenation-based feature fusion, summing the raw encoder feature maps with decoder features - which are smoothed by the centre block's dilations - lacks the flexibility to preserve smaller isolated fragments of road, as exhibited in Figure 5.

Additionally, TerraMind can narrowly outperform D-LinkNet across most pixel-based metrics, but falls short substantially in clDice and APLS. This suggests that TerraMind struggles to capture the continuous topology of roads, and potential explanations are as follows. Firstly, both baselines are based on convolutional operations that extract features with often-overlapping position-independent filters that assume spatial locality. In contrast, TerraMind projects and tokenises non-overlapping patches before the input is processed within the ViT with context from other patches incorporated through self-attention, which can flexibly attend to global information. Hence, local attention needs to be learnt. In addition, roads are characterised by the same narrow tubular structure at arbitrary locations and angles. Unlike the baselines that assume translation equivariance by design, the ViT architecture of TerraMind acquires this property through training. Consequently, the medium-sized baselines, with approximately a quarter of the parameters, outperformed TerraMind.

In Figure 6, when model performance is calculated for each level of urbanisation separately, TerraMind's urban performance is higher than the overall performance and the gap between baseline and TerraMind is smaller. Notably, TerraMind's urban APLS score exhibits a marginal difference compared to the baselines, with more rural areas, where the roads are the least dense, significantly bringing the metric down. This suggests that TerraMind struggles with sparser networks, where roads have a smaller presence and paths may not be fully visible. Although TerraMind incorporates

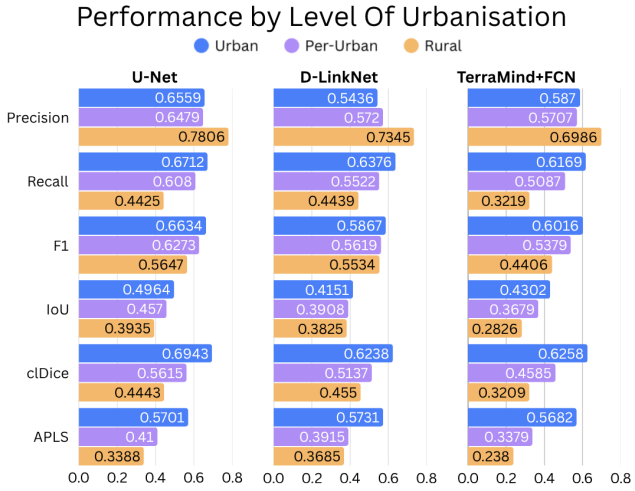


Figure 6: Mean Test-split metrics of three seeds by urbanisation level for U-Net, D-LinkNet and TerraMind+FCN

pixel-level data for spatial nuance, each image patch still contains a 160 m x 160 m area, which is projected to a coarser embedding space. Thus, the fine-grained details, such as smaller roads, may have been lost amongst the dominant background pixels before being fed into the transformers. Conversely, the model struggles to separate roads in denser networks, and learns to over-segment roads, which appear as blobs.

Rural areas also exhibit high precision and low recall. This suggests that the models have learnt to be more conservative and under-segment the areas, which is reflected in the poor APLS and clDice scores that measure connectivity. In rural areas, roads are more ambiguous, where they disappear and reappear. D-LinkNet's stronger performance on the rural roads against its counterparts could be attributed to its enhanced receptive field from its centre block of dilated convolutions. This also suggests the importance of a larger effective receptive field, and road specific mechanisms for continuity.

6 Discussion

In this project, TerraMind splits the 256 x 256 tiles into non-overlapping 16 x 16 token patches. Hence, the finest resolution available to the backbone is a 160 m x 160 m area compressed into embedding space. However, roads are thin structures and their width frequently appears narrower than one pixel in 10 m resolution satellite imagery [42]. The importance of spatial detail preserving mechanisms is a common thread amongst three notable observations. First, the multi-scale decoders (U-Net, UPerNet, DSCNet), built for spatial detail preservation, underperformed the single-resolution decoders (FCN, Linear) across most pixel-based metrics. A likely explanation is that the `LearnedInterpolateToPyramidal` neck constructs its hierarchy from tokens that differ in feature depth rather than spatial resolution [45]. Hence, the multi-scale decoders remain constrained by the 160 m granularity of the input tokens, while fusing the interpolated representations appears to obscure true road width.

Second, although TerraMind's APLS score in urban areas is comparable to the baselines, its overall score is substantially reduced by poor performance in rural areas, where roads occupy a smaller fraction of each token and background signals dominate. Third, the superior performance of U-Net (IoU = 0.4861 ± 0.0025) is attributed to it concatenation-based skip connections that give its decoder access to feature maps as fine as 20 m per pixel, making roads more discernable.

Within the decoder experiments, the FCN decoder yielded the best prediction (IoU = 0.4149 ± 0.0052) performance on all metric, except recall and clDice. Additionally, decoder capacity is observed to be less critical than how effectively a decoder leverages extracted encoder features. This is demonstrated by the Linear Decoder, which consists of a single transposed convolution, outperforming all three multi-scale decoders across pixel-based metrics while remaining within 0.012 IoU of the best decoder.

The performance across the tested LoRA configurations lags behind full fine-tuning across most metrics. Specifically, recall is preserved while precision falls by roughly eight percent, and the gap widens further for clDice and APLS, scaling directly with each metric's topological sensitivity. LoRA's failure to recover the representational capacity of full fine-tuning, evident on the training split, has two candidate explanations: the scope of adaptation, since the patch projection remains frozen, and the expressiveness of the vanilla parameterisation itself.

When benchmarked against baseline models, U-Net outperforms every TerraMind adaptation across all six metrics, despite using roughly one-quarter of the parameters. Meanwhile, TerraMind paired with an FCN decoder roughly matches D-LinkNet on pixel-based metrics, but falls substantially short on the connectivity-focused metrics, clDice and APLS.

The implications of the above results are as follows. For extracting thin structures at 10 m resolution, the U-Net remains the best option in terms of performance and resource efficiency as it reaches IoU = 0.4861 ± 0.0025 with 24.5 M parameters, a 0.294 GB checkpoint and 39 minutes of training. Although best TerraMind adaptation only narrowly falls short on the urban APLS performance compared to the baselines, the overall performance is suboptimal, with IoU = 0.4149 ± 0.0052 while using 101 M parameters, 1.22 GB and 222 minutes of training. For GFMs, if the main constraint is input resolution, and more sophisticated decoders on top of unchanged tokens will likely struggle to close the performance gap. Pathways that bypass tokenisation, and preserve finer spatial resolution, like the U-Net are more promising directions. For PEFT on GFMs, storage saving is realised. Nonetheless, further investigation is required to determine if LoRA is appropriate for recovering the full fine-tuning performance of GFMs for road extraction.

7 Conclusion

Road extraction research is mostly focused on high resolution, RGB, urban satellite data. Considering the rise of open-access medium resolution satellite data and GFMs pretrained on the data, this paper investigated whether the geospatial foundation model TerraMind can be adapted - in terms of decoder choice - to match the task-specific, medium-sized baseline models for extract roads from open-access, medium-resolution, multi-modal Sentinel-1/2 imagery

Table 4: Comparative performance of the fully fine-tuned baseline models against the Best TerraMind adaptation - fully fine-tuned TerraMind with FCN decoder - on the ROSA dataset test split.

Model	Model Param (M)	Precision	Recall	F1	IoU	clDice	APLS
U-Net [47]	24.5	**0.6578 ± 0.0047**	**0.6507 ± 0.0081**	**0.6542 ± 0.0023**	**0.4861 ± 0.0025**	**0.5752 ± 0.0029**	0.4472 ± 0.0032
D-LinkNet [60]	31.1	0.5522 ± 0.0050	0.6149 ± 0.0177	0.5818 ± 0.0056	0.4102 ± 0.0055	0.5365 ± 0.0047	**0.4506 ± 0.0074**
TerraMind + FCN Decoder	101	0.5868 ± 0.0073	0.5860 ± 0.0032	0.5864 ± 0.0052	0.4149 ± 0.0052	0.4784 ± 0.0015	0.3920 ± 0.0052

across urban, peri-urban, and rural areas. ROSA was curated in support of this aim and spans the three levels of urbanisation across the nine biomes of South Africa with 10 multispectral channels from Sentinel-2 and two radar channels from Sentinel-1, as well as government road labels.

The results corresponding to the three research questions are as follows. The best TerraMind configuration is the TerraMind with an FCN decoder, which matches the baseline models' performance most closely on urban areas, where the road signal is stronger, and falls short substantially in rural areas. No LoRA configuration effectively matched full fine-tuning performance, including performance on the training split. And, finally, fully fine-tuning U-Net with a ResNet34 backbone is the strongest model overall for medium resolution road extraction, due to its superior ability in preserving spatial detail, and the model outperforms all TerraMind adaptations across all metrics with a fourth of the parameter count and a shorter training time. Hence, it is the recommended model for settings similar to this project. The gap between TerraMind and the baseline models is attributed to the coarse 160 m x 160 m resolution image patches ingested by TerraMind in contrast to the details at 20 m resolution that the U-Net can access through its skip connections.

Additionally, a notable observation is that D-LinkNet achieves a comparable APLS score despite lagging in pixel-based metrics, reflecting a conservative prediction strategy on smaller roads that prioritises structural continuity compared to the U-Net.

7.1 Further Works

Spatial detail enhance or preservation mechanisms - such as the Fast Fourier Transform, direct image upsampling - for GFMs, like TerraMind, could be explored for road extraction on medium resolution imagery. For adaptation, a sweep under the rank-stabilised scaling factor [26] would discern whether the observed LoRA deficit is a representational capacity problem with the vanilla LoRA or a gradient collapse problem. Alternative PEFT methods, including enhanced LoRA variants such as AdaLoRA [57], could then be evaluated against that result.

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9 AI Usage Declaration

In this research, Gemini and Claude was used to generate code for visualisations, debug errors, assess experiment consistency, examine quality of the scientific method, guide the usage of the frameworks and tools, clarify content in the literature, restructure sentences, as well as proof read and evaluate paper content.

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A Supplementary Information

In this section IoU_Road refers to the same IoU defined in Equation 4 and are conducted on the TerraMind V1 Tiny Backbone with a Linear decoder and a more aggressive Early Stopping, with patience = 5.

Figure 7 displays the results of the preliminary experiments used to between using the normalisation statistic from TerraMind's pretraining and the Dataset's, as well as whether to use the enhanced RGB bands. The experiments were ran using two Kaggle T2 GPU and used trained with max epochs set at 30. The highest performing configuration was chosen for the main experiments, which was the dataset normalisation statistics and unehanced RGB channels.

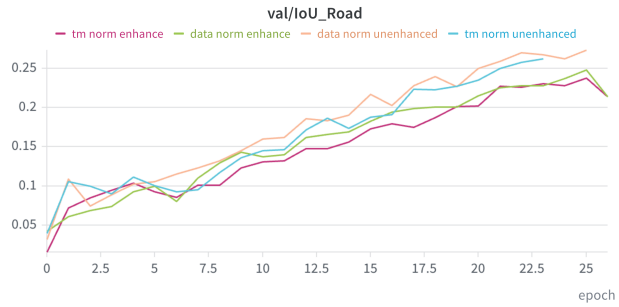


Figure 7: Preliminary experiment results using TerraMind v1 Tiny backbone with Linear decoder (tm) to decide between tm normalisation statistic (tm norm) or dataset normalisation statistic (data norm) as well as RGB enhancement (enhance or unenhanced) configuration.



Figure 8: Preliminary experiment results using TerraMind v1 Tiny backbone with Linear decoder to examine effect of lowering precision, Note that the graph titles refer to the y-axis labels and the GPU-related graphs reflect the runtime of the 10-epoch experiment

With max epoch set to 10, Figure 8 demonstrates the reduction in resource consumption when lowering the precision of model parameters, from the original 32-true to bf16-mixed, during training on L40S GPUs, without any material compromises in performance. As a result, the main experiments were conducted in bf16-mixed precision.

B Ethics

This project required no ethics clearance as it relied on open-access data, and road labels. No human interaction or personal data was collected throughout the research process and all models and software used were open-source or have an academic liscense.